

PROJECT REFERENCE GUIDE

Nano Games — Arcade Platform

Built June 17–18, 2026 · Claude Code Session · Harlem Labs

1. RESUME PROMPT — START HERE NEXT SESSION

Copy and paste this into Claude Code at the start of your next session to pick up exactly where we left off:

I'm continuing work on **Nano Games** — an arcade platform I'm building for my son Cyrus.

The project is a Next.js 16 app that serves as a hub where Cyrus can add games over time.

The first game is **Aquatic Survival** (a top-down arena game he built).

Repo: /Users/wazirigaruba/Documents/GitHub/nano-games

GitHub: github.com/WaziriG/nano-games

Vercel project: nano-games

What's built:

- Arcade lobby home page (retro pixel aesthetic, game cards, scanline effect)
- /games/aqua-survivor page with full-screen iframe + score submit modal + leaderboard
- /api/leaderboard (GET/POST) backed by Neon Postgres
- postMessage hook in aqua-survivor.html fires on game over → auto-shows score modal
- Migration script at scripts/migrate.mjs to create the leaderboard table

What still needs to be done:

- Neon Postgres database needs to be created in Vercel Storage tab and migration run
- User accounts + cloud saves (Phase 2)
- Multiplayer (Phase 3)

Help me continue building this out.

2. WHAT WAS BUILT

Overview

A Next.js 16 arcade hub deployed on Vercel. The home page is a retro-styled arcade lobby with game cards. Aquatic Survival is the first game — embedded in a full-screen iframe with a global leaderboard. Future games slot into the same structure.

File Structure

```
nano-games/ |— app/ | |— layout.tsx Root layout — loads fonts, scanline + vignette overlays | |— globals.css Tailwind 4 + retro arcade CSS (neon glow, scanlines, animations) | |— page.tsx Arcade lobby home page — game cards grid | |— games/aqua-survivor/ | | |— page.tsx Game page — iframe + score modal + leaderboard modal | | |— api/leaderboard/ | | |— route.ts GET /api/leaderboard?game=aqua-survivor | POST to submit score |— components/ | |— GameCard.tsx Reusable arcade game card (client component — has hover effects) |— lib/ | |— db.ts Neon Postgres query functions (getTopScores, insertScore) |— public/games/ | |— aqua-survivor.html The game — MODIFIED to send postMessage on game over |— scripts/ | |— migrate.mjs Run once to create the leaderboard table in Neon |— aqua-survivor.html Original standalone game (untouched — backup)
```

Key Pages

Route	What It Does
/	Arcade lobby — game cards grid, stats bar, footer. Currently shows Aquatic Survival + 2 "Coming Soon" slots.
/games/aqua-survivor	Full-screen game with top bar (back button, leaderboard, submit score). Score modal auto-appears when game over.
/api/leaderboard	GET: returns top 15 scores. POST: validates and saves a score. Requires DATABASE_URL env var.

How the Score System Works

The flow: Player finishes a run in Aquatic Survival → the game's `gameOver()` function fires a `window.parent.postMessage({ type: 'aqua-game-over', score, kills, level })` → the Next.js game page receives it → the score submission modal auto-appears → player types their name → score saved to Neon Postgres → leaderboard updates.

The score is measured in **seconds survived**. Kills and level reached are stored as secondary stats. The leaderboard sorts by seconds survived (highest first).

3. SETUP STILL NEEDED — NEON DATABASE

Status as of this session: The site is deployed on Vercel. The leaderboard API exists but the database has not been created yet. The "Top Scores" button will return an empty list until this is done.

1. **Go to Vercel** → open the nano-games project → click the **Storage** tab
2. Click **Create Database** → select **Neon Postgres** → follow the prompts. Vercel auto-adds DATABASE_URL to your environment variables.
3. **Run the migration** — go to your Neon database in the Storage tab → open the **SQL Editor** → paste and run:

```
CREATE TABLE IF NOT EXISTS leaderboard ( id SERIAL PRIMARY KEY, game_slug VARCHAR(50) NOT NULL, player_name VARCHAR(30) NOT NULL, score INTEGER NOT NULL, kills INTEGER NOT NULL DEFAULT 0, level INTEGER NOT NULL DEFAULT 1, created_at TIMESTAMPTZ NOT NULL DEFAULT NOW() ); CREATE INDEX IF NOT EXISTS leaderboard_game_score_idx ON leaderboard (game_slug, score DESC);
```

1. **Redeploy** the Vercel project so it picks up the new DATABASE_URL — Vercel → Deployments → Redeploy latest.

4. HOW TO ADD A FUTURE GAME

The arcade is designed so Cyrus can drop in new games over time. Here's the pattern:

1. **Add the game file** — drop the HTML game file into public/games/ (e.g., public/games/space-blast.html)
2. **Add a card on the home page** — open app/page.tsx and add a new entry to the GAMES array with live: true and the new slug/colors/description
3. **Create a game page** — create app/games/space-blast/page.tsx (copy the aqua-survivor page and change the game name and iframe src)
4. **Enable leaderboard for new game** — in app/api/leaderboard/route.ts, add the new slug to the VALID_GAMES set
5. **Wire up postMessage** — add window.parent.postMessage({ type: 'GAME-SLUG-game-over', score, kills, level }, '*') to the game's "game over" function, then update the event listener in the new page

No database changes needed for new games — the leaderboard table stores `game_slug` so all games share the same table automatically.

5. TECH STACK

Layer	Technology	Notes
Framework	Next.js 16 (App Router)	Server components by default; game page is client component
Language	TypeScript 5 (strict)	No any — use unknown and narrow
Styling	Tailwind CSS 4	CSS-first config; custom arcade tokens in <code>globals.css</code>
Database	Neon Postgres	Via Vercel Storage — <code>@neondatabase/serverless</code> driver
Hosting	Vercel	Auto-deploys on every push to <code>main</code>
Game engine	Vanilla JS + HTML5 Canvas	Single-file, no dependencies — Cyrus built it himself

6. ROADMAP — WHAT'S NEXT

Phase	Feature	Complexity	Notes
Now	Set up Neon database (see Section 3)	Low	15 min in Vercel UI
Phase 2	User accounts + cloud saves	Medium	Add auth (Clerk or simple JWT) + save <code>localStorage</code> state to Postgres
Phase 2	Cyrus's profile page	Low	Show his games, his high scores, a bio
Done	Custom domain	Low	Live at <code>nanogames.app</code> — the arcade was rebranded from Cys Gaming World to Nano Games
Phase 3	Multiplayer	High	

Phase	Feature	Complexity	Notes
			Needs WebSocket server (Vercel Durable Objects or Cloudflare) — separate infrastructure conversation
Anytime	Add a 2nd game	Low	Follow the pattern in Section 4

7. KEY LINKS & REFERENCES

Resource	Location
Local code	<code>/Users/wazirigaruba/Documents/GitHub/nano-games</code>
GitHub repo	<code>github.com/WaziriG/nano-games</code>
Vercel project	<code>nano-games</code> (in Waziri Garuba's projects)
Game file (live)	<code>public/games/aqua-survivor.html</code>
Game file (original)	<code>aqua-survivor.html</code> (root — untouched backup)
Leaderboard API	<code>/api/leaderboard?game=aqua-survivor</code>